Huffman Codes

Encoding messages

- Encode a message composed of a string of characters
- Codes used by computer systems
  - ASCII
    - uses 8 bits per character
    - can encode 256 characters
  - Unicode
    - 16 bits per character
    - can encode 65536 characters
    - includes all characters encoded by ASCII
- ASCII and Unicode are *fixed-length codes*
  - all characters represented by same number of bits

Problems

- Suppose that we want to encode a message constructed from the symbols **A**, **B**, **C**, **D**, and **E** using a fixed-length code
  - How many bits are required to encode each symbol?
    - at least 3 bits are required
    - 2 bits are not enough (can only encode four symbols)
  - How many bits are required to encode the message **DEAACAAAAABA**?
    - there are twelve symbols, each requires 3 bits
    - 12*3 = 36 bits are required

Drawbacks of fixed-length codes

- Wasted space
  - Unicode uses twice as much space as ASCII
    - inefficient for plain-text messages containing only ASCII characters
- Same number of bits used to represent all characters
  - ‘a’ and ‘e’ occur more frequently than ‘q’ and ‘z’

- **Potential solution**: use variable-length codes
  - variable number of bits to represent characters when frequency of occurrence is known
  - short codes for characters that occur frequently

Advantages of variable-length codes

- The advantage of variable-length codes over fixed-length is short codes can be given to characters that occur frequently
  - on average, the length of the encoded message is less than fixed-length encoding
- **Potential problem:** how do we know where one character ends and another begins?
    - not a problem if number of bits is fixed!

A = 00
B = 01
C = 10
D = 11

0010110111001111111111

A C D B A D D D D D

Prefix property

- A code has the **prefix property** if no character code is the prefix (start of the code) for another character
- Example:

Symbol	Code
P	000
Q	11
R	01
S	001
T	10

01001101100010

R S T Q P T

- 000 is not a prefix of 11, 01, 001, or 10
- 11 is not a prefix of 000, 01, 001, or 10 …

Code without prefix property

- The following code does **not** have prefix property

Symbol	Code
P	0
Q	1
R	01
S	10
T	11

- The pattern **1110** can be decoded as **QQQP**, **QTP**, **QQS**, or **TS**

Problem

- Design a variable-length prefix-free code such that the message **DEAACAAAAABA** can be encoded using 22 bits
- Possible solution:
  - **A** occurs eight times while **B**, **C**, **D**, and **E** each occur once
  - represent **A** with a one bit code, say 0
    - remaining codes cannot start with 0
  - represent **B** with the two bit code 10
    - remaining codes cannot start with 0 or 10
  - represent **C** with 110
  - represent **D** with 1110
  - represent **E** with 11110

Encoded message

DEAACAAAAABA

Symbol	Code
A	0
B	10
C	110
D	1110
E	11110

1110111100011000000100 22 bits

Another possible code

DEAACAAAAABA

Symbol	**Code**
A	0
B	100
C	101
D	1101
E	1111

1101111100101000001000 22 bits

Better code

DEAACAAAAABA

Symbol	**Code**
A	0
B	100
C	101
D	110
E	111

11011100101000001000

20 bits

What code to use?

- Question: Is there a variable-length code that makes the most efficient use of space?

Answer: Yes!

Huffman coding tree

- Binary tree
  - each leaf contains symbol (character)
  - label edge from node to left child with 0
  - label edge from node to right child with 1
- Code for any symbol obtained by following path from root to the leaf containing symbol
- Code has prefix property
  - leaf node cannot appear on path to another leaf
  - *note*: fixed-length codes are represented by a complete Huffman tree and clearly have the prefix property

Building a Huffman tree

- Find frequencies of each symbol occurring in message
- Begin with a forest of single node trees
  - each contain symbol and its frequency
- Do recursively
  - select two trees with smallest frequency at the root
  - produce a new binary tree with the selected trees as children and store the sum of their frequencies in the root
- Recursion ends when there is one tree
  - this is the Huffman coding tree

Example

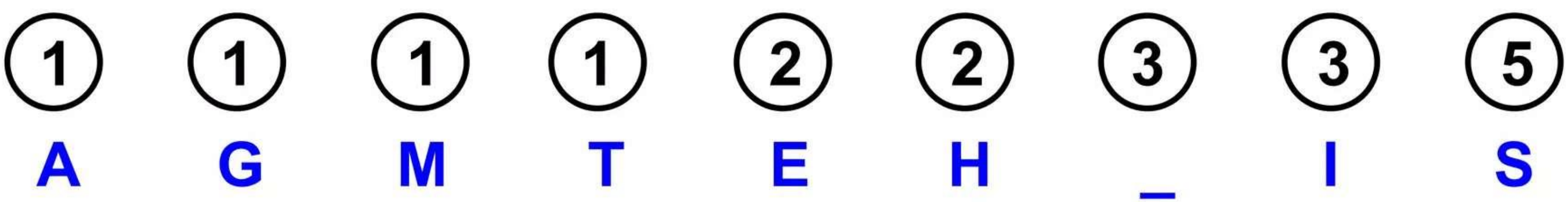
- Build the Huffman coding tree for the message

This is his message

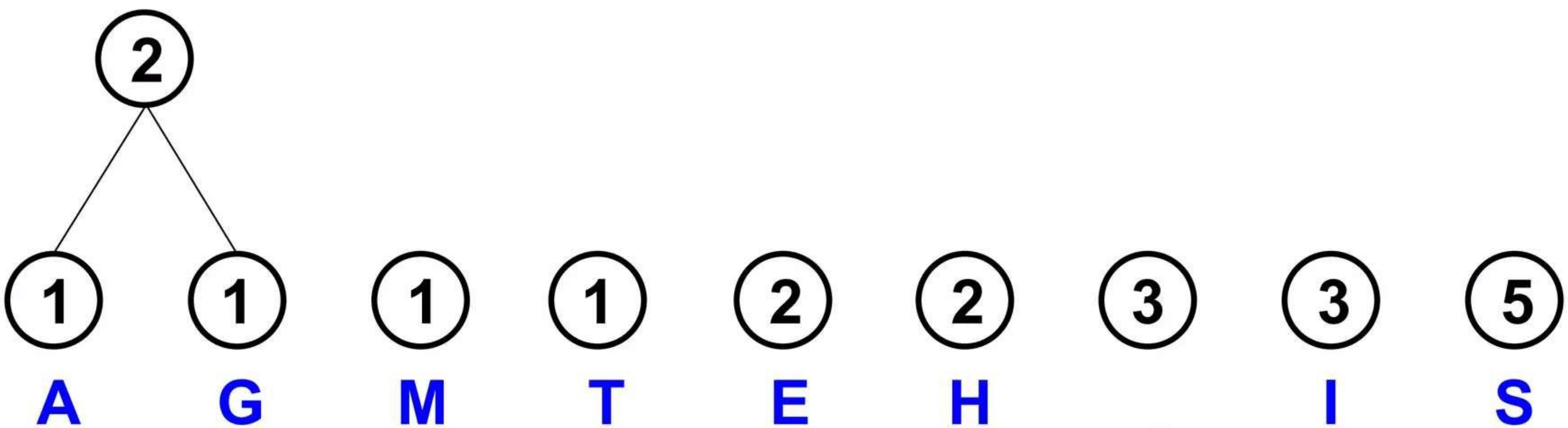
- Character frequencies

A	G	M	T	E	H	_	I	S
1	1	1	1	2	2	3	3	5

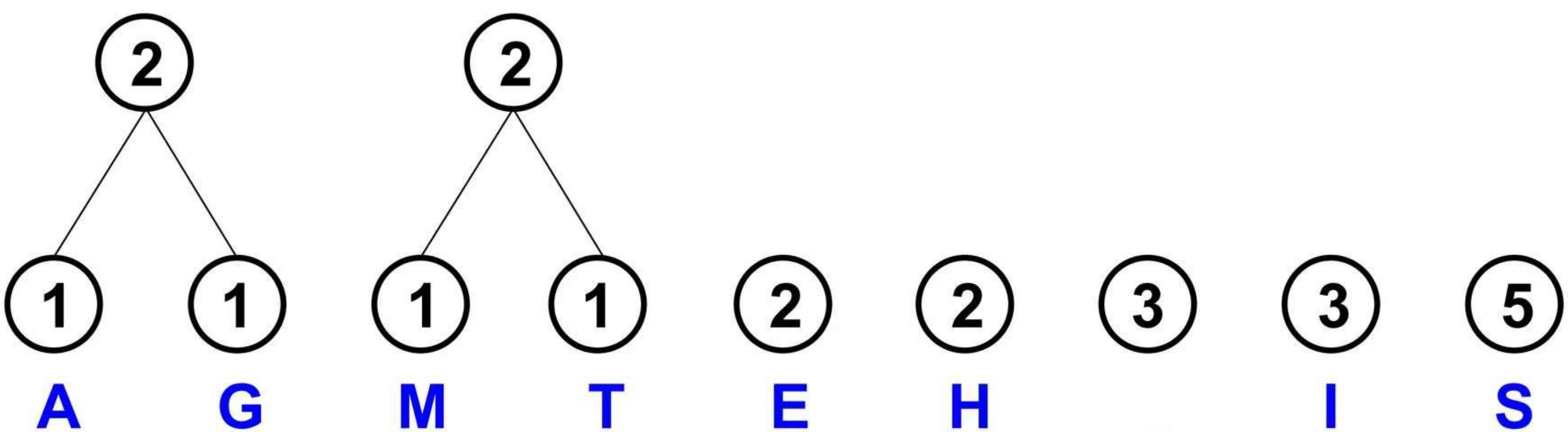
- Begin with forest of single trees



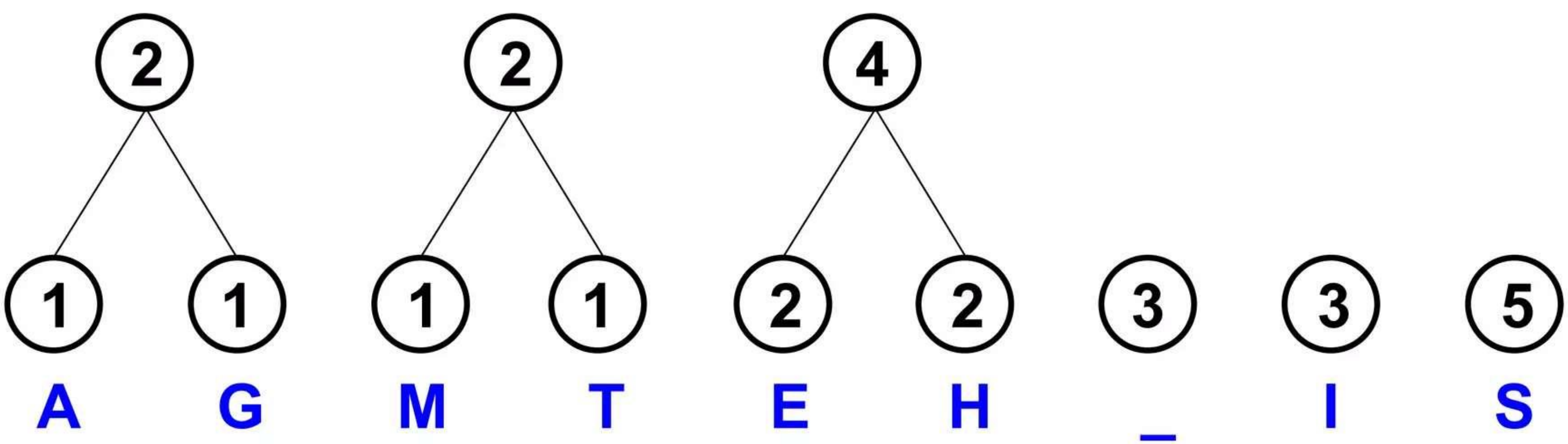
Step 1



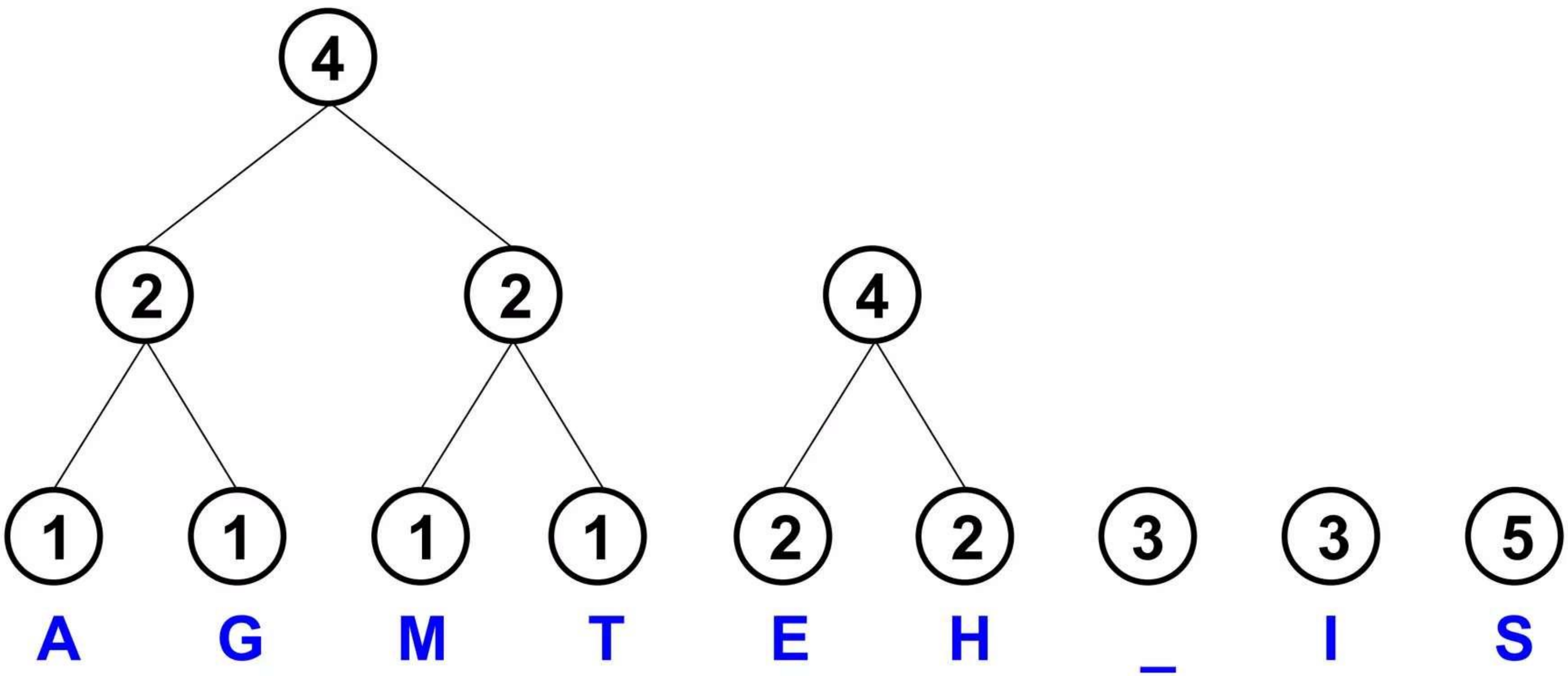
Step 2



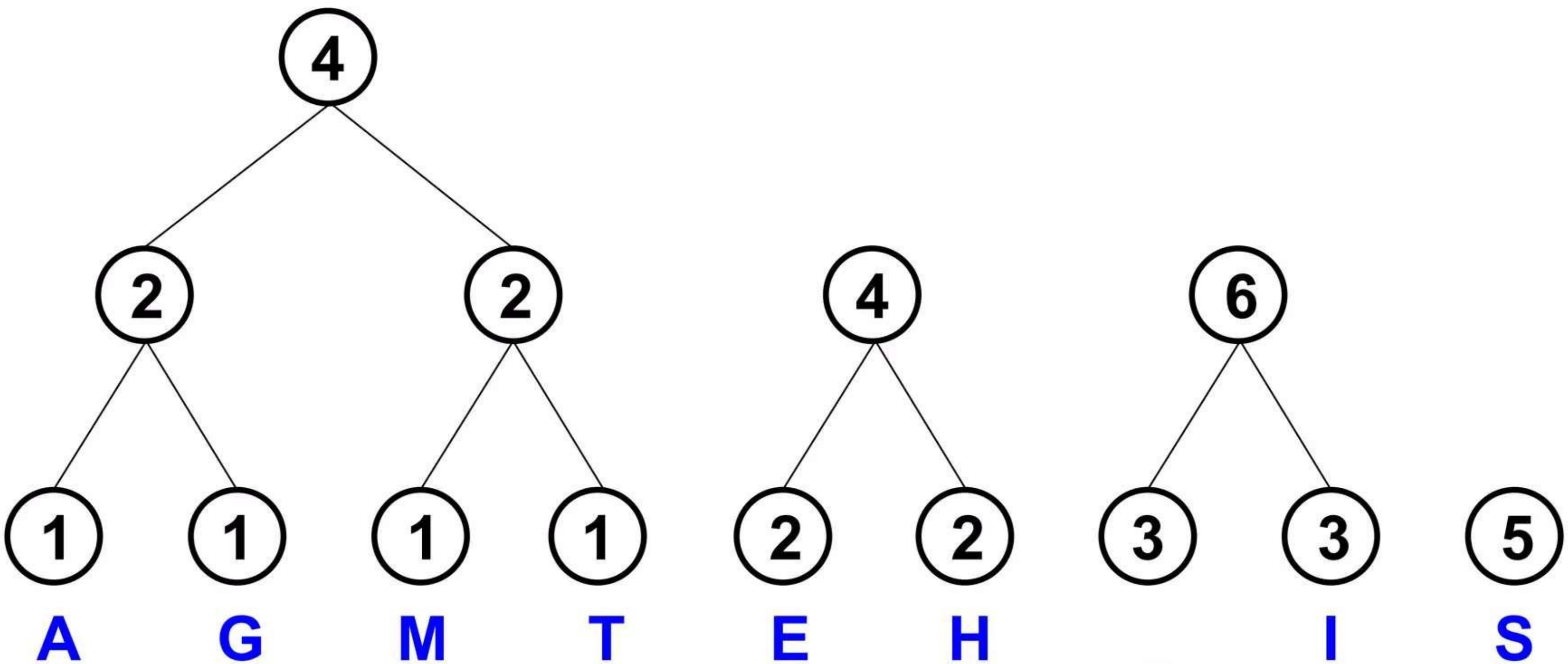
Step 3



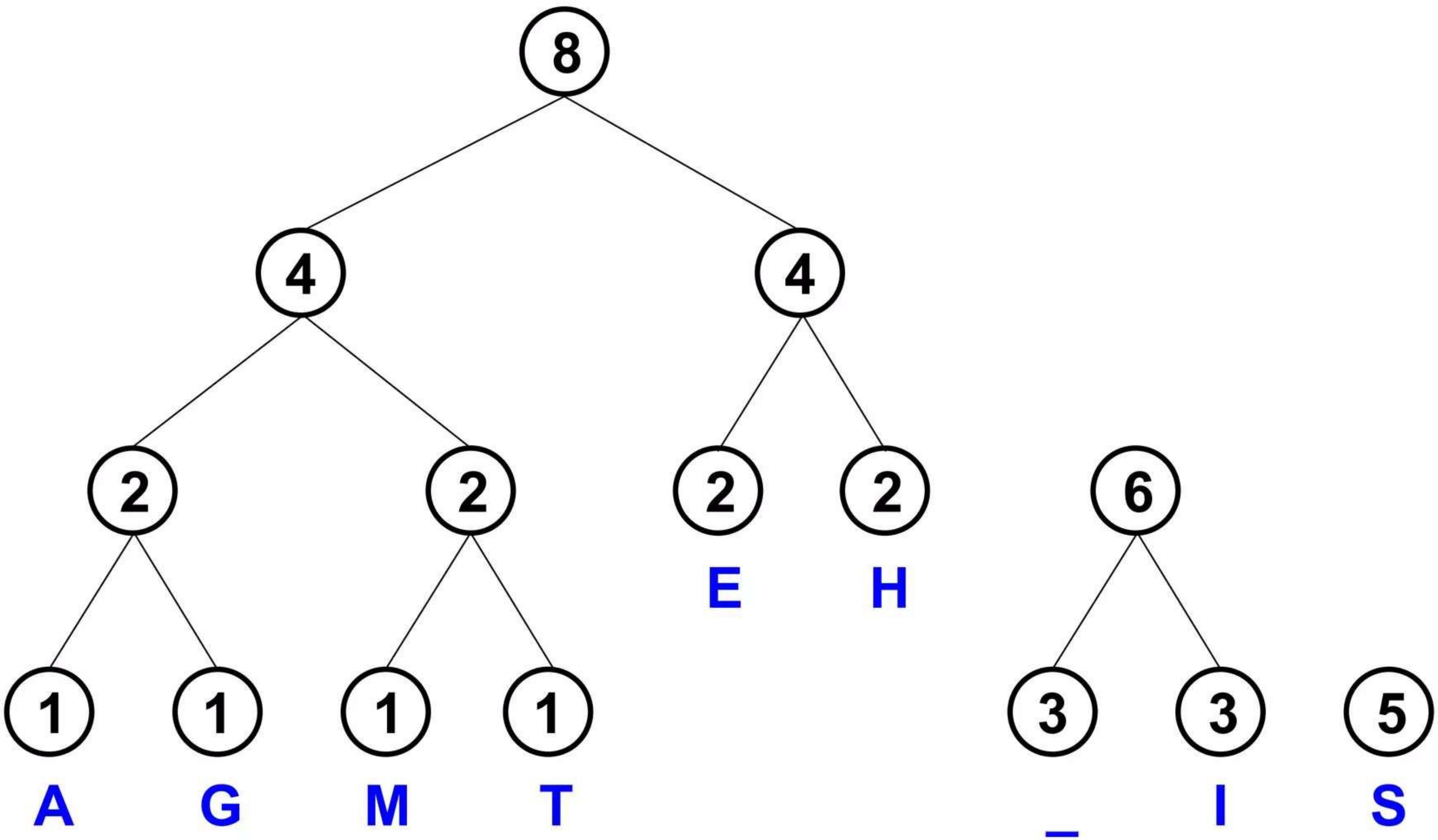
Step 4



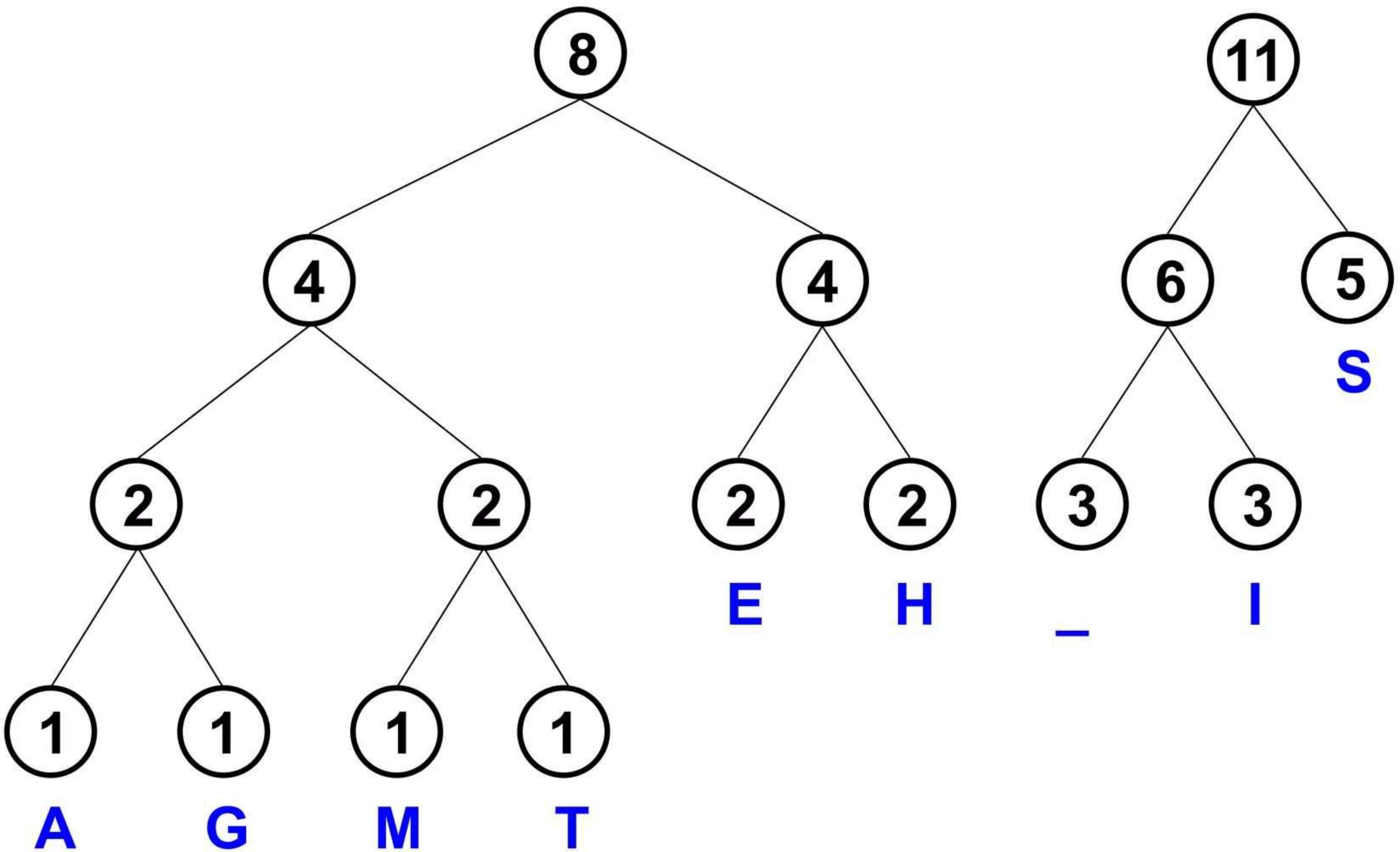
Step 5



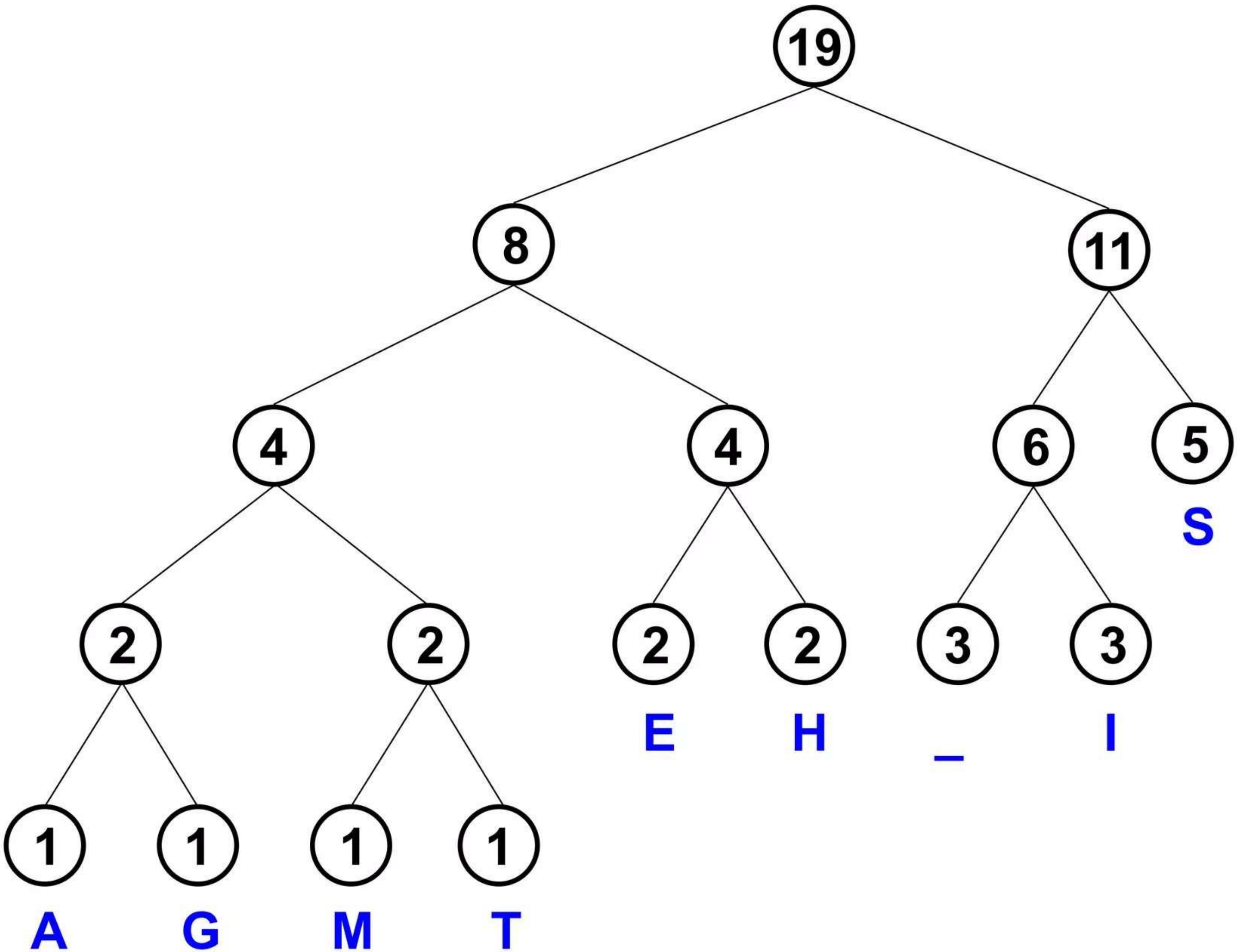
Step 6



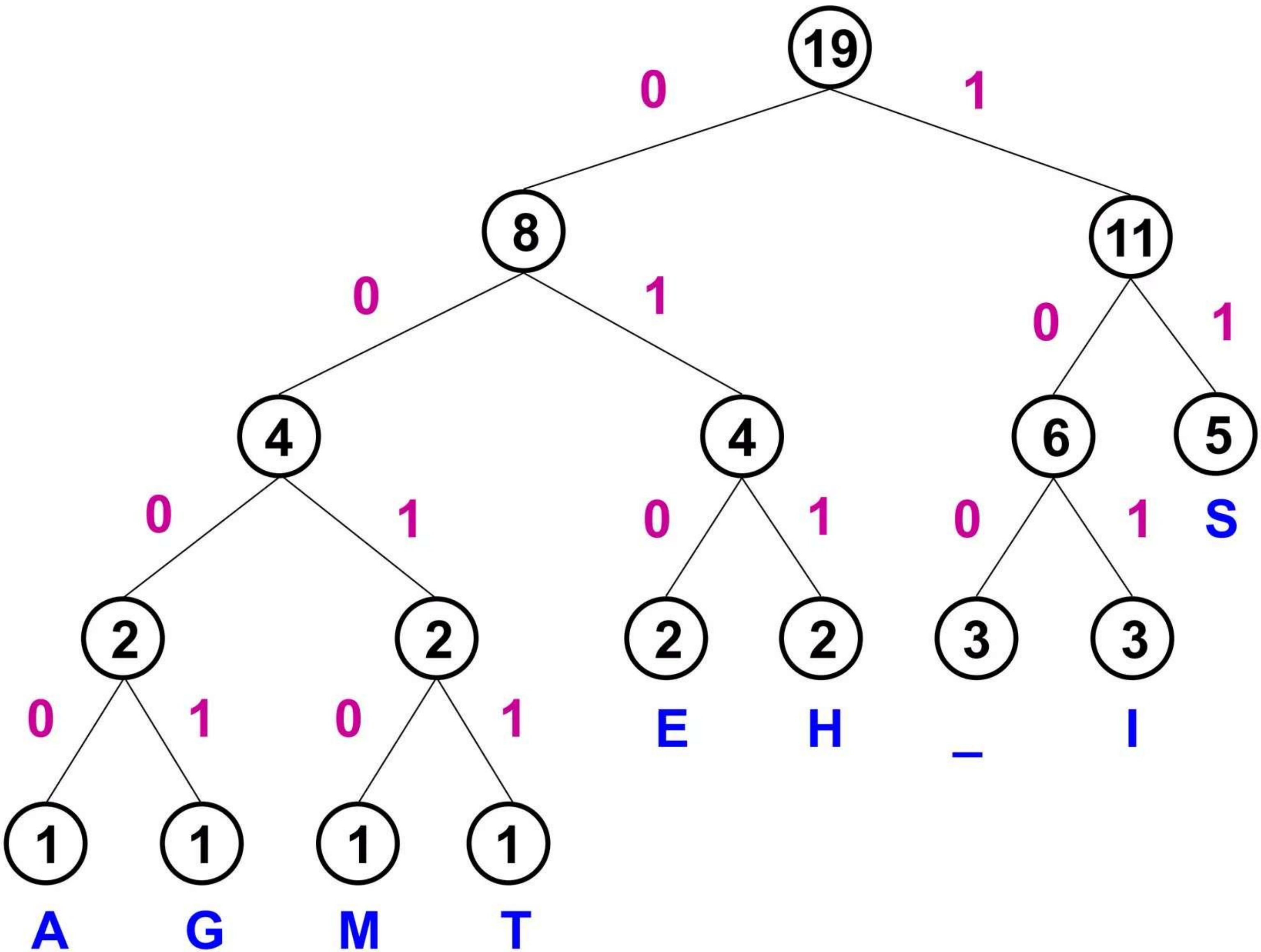
Step 7



Step 8

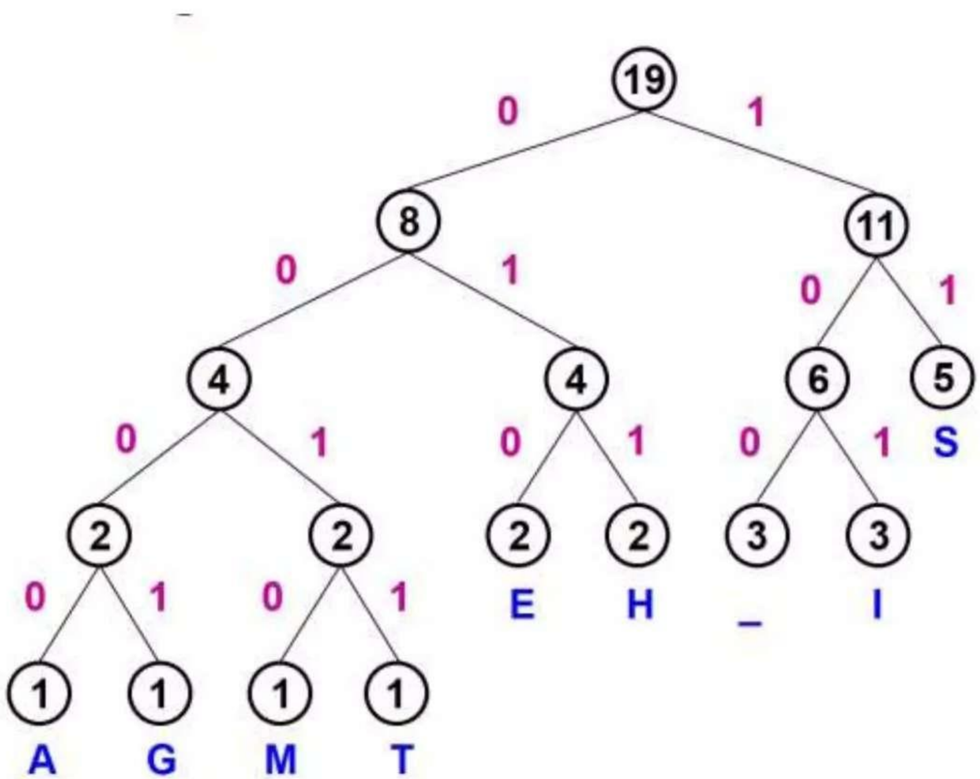


Label edges



Huffman code & encoded message

This is his message



S	11
E	010
H	011
_	100
I	101
A	0000
G	0001
M	0010
T	0011

0011011101111001011110001110111100001001011110000000001010

Average Code Length

The average code length of the Huffman tree can be determined by using the formula given below:

Average Code Length = ∑ (frequency × code length) / ∑ (frequency)

This is his message

Symbol	Frequency (F)	Code	Code Length (CL)	Total (F*CL)
S	5	11	2	10
E	2	010	3	6
H	2	011	3	6
_	3	100	3	9
I	3	101	3	9
A	1	0000	4	4
G	1	0001	4	4
M	1	0010	4	4
T	1	0011	4	4
Total_Freq	**19**			**56**

Average = 56/19
= 2.94737 bits

Length of the string??